

इंटरनेट

मानक

Disclosure to Promote the Right To Information

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“जानने का अधिकार, जीने का अधिकार”

Mazdoor Kisan Shakti Sangathan

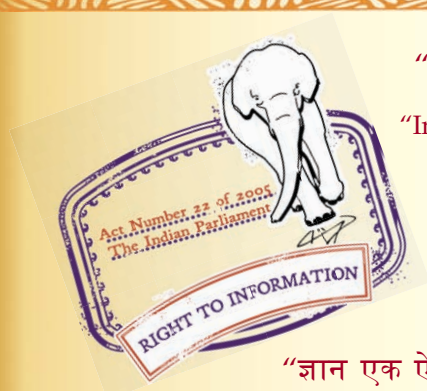
“The Right to Information, The Right to Live”

“पुराने को छोड़ नये के तरफ”

Jawaharlal Nehru

“Step Out From the Old to the New”

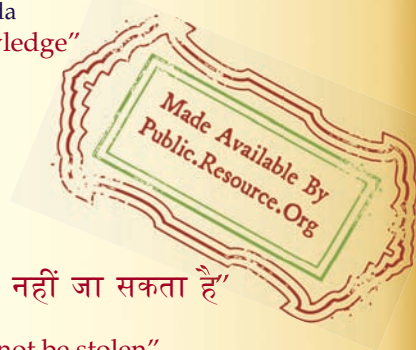
IS 1417 (1999): Gold and Gold Alloys, Jewellery/Artefacts -
Fineness and Marking [MTD 10: Precious Metals]



“ज्ञान से एक नये भारत का निर्माण”

Satyanarayan Gangaram Pitroda

“Invent a New India Using Knowledge”



“ज्ञान एक ऐसा खजाना है जो कभी चुराया नहीं जा सकता है”

Bhartrhari—Nitiśatakam

“Knowledge is such a treasure which cannot be stolen”

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भारतीय मानक

REAFFIRMED

2004

स्वर्ण एवं स्वर्ण मिश्रधातुएं, आभूषण/शिल्पकारी शुद्धता एवं
मुहरांकन — विशिष्टि
(तीसरा पुनरीक्षण)

Indian Standard
GOLD AND GOLD ALLOYS, JEWELLERY/
ARTEFACTS — FINENESS AND MARKING —
SPECIFICATION
(*Third Revision*)

ICS 39.060

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BUREAU OF INDIAN STANDARDS
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NEW DELHI 110002

FOREWORD

This Indian Standard (Third Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Precious Metals Sectional Committee had been approved by the Metallurgical Engineering Division Council.

This standard was first published in 1959 and subsequently revised in 1971 and 1981. In many advanced countries, the gold ornaments/artefacts traded are certified for purity and are marked with various details like purity/fineness of the ornament, manufacturer's mark, the mark of the assaying laboratory responsible for certifying the purity etc. These ornaments/artefacts are termed as Hallmarked. The need for availability of jewellery of certified purity in the country was the major reason for undertaking the present revision which is an amalgamated revision of IS 1417 and provisions of IS 8844 : 1978 'Guidelines for marking purity of gold on gold articles/ornaments' have been incorporated after suitable modifications. Consequent upon publication of this standard IS 8844 shall stand withdrawn.

This standard is primarily meant to classify the various grades of gold according to their gold content and provide guidelines to mark the gold articles. The percentage of alloying elements stipulated for various grades depend upon their specific uses and the detailed compositions of these grades have been covered in IS 2790 : 1999 'Guidelines for manufacture of 23, 22, 21, 18, 14 and 9 carat gold alloys (*second revision*)'.

In this revision, the grades of fineness 500 and 970 have been deleted and a new grade of fineness 958 has been introduced on account of its popularity for manufacture of ornaments/artefacts in many parts of the country.

On the subject also exists an International Standard ISO 9202 : 1991 'Jewellery — Fineness of precious metal alloys' specifying the finenesses of alloys of various precious metals like Pt, Pd and Ag alloys apart from gold. The fineness of gold and gold alloys specified in this standard are totally in line with the fineness for gold alloys specified in ISO standard thereby aligning Indian Standard with the provisions of international standard. The standard also specifies additional grades gold alloys used in India.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

AMENDMENT NO. 1 DECEMBER 2004
TO
IS 1417 : 1999 GOLD AND GOLD
ALLOYS, JEWELLERY/ARTEFACTS — FINENESS AND
MARKING — SPECIFICATION

(Third Revision)

(Page 1, clause 4.1, under notes) — Add the following notes:

3 Gold alloys for manufacturing jewellery/artefacts and gold solders shall be free from iridium and ruthenium.

4 Presence of ruthenium, iridium and cadmium in gold and gold alloys shall be detected by X-ray fluorescence (XRF) method till an established test procedure is formulated for the determination of these elements.'

AMENDMENT NO. 2 DECEMBER 2007
TO
IS 1417 : 1999 GOLD AND GOLD ALLOYS,
JEWELLERY/ARTEFACTS — FINENESS AND
MARKING — SPECIFICATION
(Third Revision)

(Page 1, clause 1.2) — Insert the following note at the end of the clause:

‘NOTE – Artefacts cover medallions, utensils and coins provided the purity of gold alloy for artefacts is as per 4.’

(Page 1, clause 4.1) — Insert the following new grade 708 in between 750 and 585 as follows:

<i>Grade</i>	<i>Fineness, Min</i>	<i>Carat</i>
708	708	17

[Page 1, clause 4.1, Note (see also Amendment No. 1)] — Insert the following new note at the end:

‘5 For Jewellery/Artefacts of fineness 995 and 999, the minimum fineness of 995 and 999 shall be ensured irrespective of the solder used.’

(Page 2, clause 5.2) — Insert the following note at the end of the clause:

‘NOTE – The requirement for solder of jewellery of fineness 995 and 999 shall be as per agreement between manufacturer and purchaser until the same has been specified in IS 3095. However, solders used shall be free from cadmium.’

(Page 2, clause 6.1, lines 4 and 5) — Substitute ‘hallmarking centres only. The hallmarked articles shall carry the following markings:’ for ‘hallmarking centres only which comprise of the following:’

[Page 2, clause 6.1(a) to (c)] — Substitute the following for the existing:

- ‘a) BIS Mark
- b) Purity grade/fineness
- c) Assay Centre’s Identification Mark
- d) Year of marking denoted by a letter symbol (as defined by BIS);
and
- e) Jeweller’s mark/manufacturer’s identification mark.

Amend No. 2 to IS 1417 : 1999

Hallmarking — The use of Hallmark is governed by the provisions of the *Bureau of Indian Standards Act, 1986* and Rules and Regulations made thereunder. The details of conditions under which the licence for the use of the Standard Mark may be granted to jewellers/jewellery manufacturers may be obtained from Bureau of Indian Standards.'

(Page 2, clause 6.1, Note) — Delete the complete note.

(Page 2, clause 6.1.3) — Insert the following new clause after 6.1.3:

'6.1.4 Marking to next lower fineness/caratage as given in the standard to be done when the gold article/ornament fails to comply with specified fineness/caratage.'

(Page 2, clause 6.2) — Substitute the following for the existing clause:

6.2 Gold jewellery/artefacts on which marking should not be applied are as follows:

- a) Any gold jewellery/artefacts which has not been assayed by BIS recognized Assaying and Hallmarking Centres.
- b) Any gold article/ornament which is not solid or which is made with a hollow center and then filled with base metal, cement, lac or other foreign substance thereby giving the article a false or added weight or strength unless the weight of gold is declared by the jeweller and marked on the article by the Centre.
- c) Gold article/ornament on which it is physically impossible to stamp the marks.
- d) Below 2 g gold.
- e) Below 9 carat gold.

(MTD 10)

AMENDMENT NO. 3 JUNE 2010
TO
IS 1417 : 1999 GOLD AND GOLD ALLOYS,
JEWELLERY/ARTEFACTS — FINENESS AND
MARKING — SPECIFICATION

(Third Revision)

[Page 2, clause 6.1.4 (see also Amendment No. 2)] — Delete clause 6.1.4.

[Page 2, clause 6.2(d) (see also Amendment No. 2)] — Delete.

(MTD 10)

Reprography Unit, BIS, New Delhi, India

Indian Standard

GOLD AND GOLD ALLOYS, JEWELLERY/ ARTEFACTS — FINENESS AND MARKING — SPECIFICATION (*Third Revision*)

1 SCOPE

1.1 This standard specifies eight grades of gold, used in the manufacture of jewellery/artefacts of gold, based on their gold content.

1.2 This standard also specifies the guidelines for marking of purity and other details on tested jewellery/artefacts.

2 REFERENCES

The following standards contain provisions which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent edition of the standard indicated below:

<i>IS No.</i>	<i>Title</i>
1418 : 1999	Assaying of gold in gold bullion, gold alloys and gold jewellery/artefacts — Cupellation (fire assay) method (<i>second revision</i>)
2790 : 1999	Guidelines for manufacture of 23, 22, 21, 18, 14 and 9 carat gold alloys (<i>first revision</i>)
3095 : 1999	Gold solders for use in manufacture of jewellery — Specification (<i>second revision</i>)

3 TERMINOLOGY

For the purpose of this standard, the following definitions shall apply.

3.1 Assaying

The method of accurate determination of the gold content of the sample expressed in parts per thousand (‰).

3.2 Carat

One-twenty fourth part by mass of the metallic element gold.

3.3 Fineness

The ratio between the mass of gold content and the total mass expressed in parts per thousand (‰).

3.4 Fine Gold

It is gold having fineness 999 parts per thousand (‰) and above without any negative tolerance.

3.5 Gold

The metallic element gold, free from any other element.

3.6 Gold Alloy

Gold alloyed with one or more other elements.

3.7 Standard Gold

Gold having fineness 995 parts per thousand (‰) and above without any negative tolerance.

4 GRADES

4.1 Gold and gold alloys shall be classified in accordance with the following grades depending upon their fineness.

<i>Grade</i>	<i>Fineness, Min</i>	<i>Carat</i>
Fine gold	999	—
Standard gold	995	—
958.3	958	23
916.6	916	22
875	875	21
750	750	18
585	585	14
375	375	9

NOTES

1 The above classification is applicable for gold jewellery/artefacts also.

2 For jewellery/artefacts of 23 carat, the gold solder of 22 grade may be used. The purity of the grade of gold alloy may be suitably adjusted to make the melting purity to the declared fineness.

5 REQUIREMENTS

5.1 The fine or standard gold used for the manufacture of jewellery shall have the fineness as agreed to

between the purchaser and the manufacturer subject to a minimum of fineness as given in 4.1 without any negative tolerance.

5.2 Solders used for the manufacture of gold jewellery/artefacts shall be of the same fineness (or caratage) as that of the jewellery/ article and shall conform to the requirements as specified in IS 3095. However in case of 23 carat jewellery/artefacts where it is not possible to use solder of the same caratage, solders of immediate next lower caratage may be used provided the overall purity of the jewellery as declared conforms to the respective grade and fineness. For 9, 14, 18, 21 and 22 carat jewellery/artefacts solders shall be of the same caratage as the jewellery article. Solder compositions other than specified in IS 3095 may also be used provided their gold content is same as the respective grade as specified for the standard solder and they are free from cadmium. Solders of a fineness less than the standard fineness of article shall not be used for strengthening, weighing or filing.

5.3 Sampling

Unless otherwise specified for the jewellery/article of the same design and quality, the sample should be taken from 10 percent of the articles/ornaments selected at random. In all other cases, the sample should be taken from each article/jewellery. Any other sampling plan may also be followed subject to agreement between the contracting parties or as specified by the inspection agency as the case may be.

5.4 Assaying

The sample taken as prescribed in 5.3 shall be assayed in accordance with IS 1418. The actual weight of samples etc shall however be as prescribed in respective test method detailed in IS 1418.

5.5 Tolerances

5.5.1 No negative tolerance shall be allowed on the gold content of gold article/jewellery sampled and tested as per 5.3.

6 MARKING

6.1 The gold, gold alloys, jewellery/artefacts shall be stamped with the Standard Mark in this case known

as the "Hallmark" by BIS recognized assaying and hallmarking centers only which comprise of the following:

- a) Assay center's symbol;
- b) The purity mark along with purity/fineness grade (indicating caratage shall not be mandatory); and
- c) Year of marking denoted by a letter symbol (as defined by BIS);

Hallmarking — The use of Hallmark is governed by the provisions of the *Bureau of Indian Standards Act, 1986* and Rules and Regulations made thereunder. The details of conditions under which the licence for the use of the Standard Mark may be granted to jewellers/jewellery manufacturers may be obtained from Bureau of Indian Standards.

NOTE — Jeweller's/Sponsor's logo on each article shall be marked before offering the lot to BIS recognized Assaying and Hallmarking Centre for Hallmarking.

6.1.1 Only linear designs shall be permitted.

6.1.2 Marking shall be legible and durable.

6.1.3 Marking to be done on all parts which can be easily removed or replaced. In case of bangles and light items marking shall be done at one place only.

6.2 Gold jewellery/artefacts on which marking should not be applied are as follows :

- a) Any gold jewellery/artefacts which has not been assayed by BIS recognized Assaying and Hallmarking Centers.
- b) Any gold article/ornament which fails to comply with specified fineness/caratage.
- c) Any gold article/ornament which is not solid or which is made with a hollow center and then filled with base metal, cement, lac or other foreign substance thereby giving the article a false or added weight or strength unless the weight of gold is indicated on the article along with the jeweller's/ sponsor's logo.
- d) Gold article/ornament on which it is physically impossible to stamp the marks.

Bureau of Indian Standards

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Amendments are issued to standards as the need arises on the basis of comments. Standards are also reviewed periodically; a standard along with amendments is reaffirmed when such review indicates that no changes are needed; if the review indicates that changes are needed, it is taken up for revision. Users of Indian Standards should ascertain that they are in possession of the latest amendments or edition by referring to the latest issue of 'BIS Catalogue' and 'Standards: Monthly Additions'.

This Indian Standard has been developed from Doc : No. MTD 10 (4277).

Amendments Issued Since Publication

Amend No.	Date of Issue	Text Affected

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